Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A Bio-Semiconductor–Based Point-of-Care Multimodal Biosensing Platform for Osteoporosis Progression Diagnosis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio & Medicine			
	Medium Category	Small Category	Medium Category	Small Category
	e.g. Precision manufacturing machinery	e.g. Cutting/processing machinery		
Concept of Proposed Technology	• A nanobody will serve as a highly selective and highly reactive bioreceptor, immobilized on nanopatterned electrodes to enhance biosensing performance. • Using a semiconductor fabrication process to obtain cheap, scalable, reproducible, and high-performance nanopatterned electrodes. • The proposed immune biosensing platform can detect extremely low concentrations of biomarkers, enabling early-stage disease or disease progression over time. • Multimodal biosensing platform to improve accuracy and reliability			
Rationale for Joint R&D Project	To improve biosensor performance for diagnosing osteoporosis and its progression over time, bioreceptors that recognize only target biomarkers within complex biological fluids (e.g., blood or serum, which contain high concentrations of non-specific biomolecules) without significantly increasing resistance, and their alignment, are required. Therefore, there is a need to collaborate with companies that can develop highly sensitive and reactive bioreceptors and nanopatterned electrodes to improve bioreceptor alignment and inhibit the immobilization of nonspecific biomolecules on the electrode, thereby significantly improving biosensing performance.			
Key Words	1. Nanobody-based Electrochemical biosensor 2. Semiconductor-based nanopatterned electrodes			

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	3. Rapid Point-of-care system 4. Multimodal biosensing platform to improve accuracy and reliability 5. Osteoporosis and its progression diagnostics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	Develop Bio-Semiconductor–Based Point-of-Care Multimodal Biosensing Platform for Osteoporosis Diagnosis			
R&D Target Specifications	• Bio-semiconductor–based POC multimodal duplex platform (electrochemical + ELISA) for osteoporosis assessment (i.e., P1NP/CTx) • Multiplex quantification: ≥2 biomarkers in serum/plasma, ≤50 μL • Sensitivity: ng/mL-level detection, LOD ≤ 1 ng/mL • Analytical performance: CV ≤ 15% • Product format: portable cartridge + reader • Speed & usability: time-to-result ≤ 20 min, ≤3-step workflow • Reliability: built-in internal QC for field use			
R&D Plan	**Aim 1. Develop high-performance biosensors capable of detecting multiplexed biomarkers (0 month – 18 months).** 1.1: Design and development of effective antigens for high-selectivity and high-reactivity antibody production for biosensor applications • Nanobody engineering based on identified antibody gene sequences • Screening and characterization of high-Affinity and high-stability nanobodies for biosensing applications • Evaluate batch-to-batch variation and reproducibility 1.2: Develop semiconductor fabrication methods for mass-produced and reproducible nanopatterned electrodes • Develop and optimize semiconductor processes to obtain well-defined nanopatterned electrodes on a wafer larger than 6” • Characterization of obtained nanopatterned electrodes for electrical and electrochemical properties • Characterization of the size of nanopattern and electrical and electrochemical properties of nanopatterned electrodes with various batches for confirming reproducibility of electrodes 1.3: Develop immobilization protocols of nanobodies on the nanopatterned electrodes and rapid detection protocols • Study to define self-assembly monolayer and linkers to increase the number of nanobody and improve alignment of nanobody and stable immobilization on the nanopattern • Design and develop a rapid detection system using electrically active small molecules • Optimize blocking buffer to avoid non-specific biomolecules immobilization in complex biological solution 1.4: Develop highly sensitive and selective nanobodies-based biosensors capable of detecting multiplexed biomarkers for monitoring of osteoporosis disease progression • Assemble self-assembly monolayer, nanobody, and electrically active small molecules on the nanopatterned electrode and optimize incubation time • Evaluate biosensing performance with two targeted biomarkers and analyze data for LOD, sensitivity, selectivity, and reproducibility **Aim 2. Develop portable and multimodal POC diagnostic platform (18 months – 28 months)** 2.1: Establish multimodal and multiplexed POC diagnostic platform • Develop an ELISA detection system using the well-defined nanobody and characterize its detection range, LOD, sensitivity, selectivity, and reproducibility • Develop two nanopatterned electrode arrays to achieve multiplexed biomarker detection on a single chip and one-time detection			

	• Evaluate biosensing performance with two targeted biomarkers, analyze data for LOD, sensitivity, selectivity, and reproducibility, and then compare the obtained results with the ELISA assay 2.2: Validation of prototype the POC kit in the laboratory using biological samples • Establish a prototype of multimodal POC kit • Validation of two biomarkers in serum or blood and evaluate biosensing performance with electrochemical biosensor and ELISA kit **Aim 3. Pre-clinical validation (29 months – 36 months)** 3.1: Validate cross-reactivity with different disease and reproducibility • Validate cross-reactivity of the POC kit using samples containing two targeted biomarkers, along with hormones and major biomarkers associated with endocrine diseas (Osteoporosis is a disease that weakens bones by reducing bone density. It is often accompanied by endocrine diseases such as diabetes, thyroid disease (thyrotoxicosis), hyperparathyroidism, and Cushing's syndrome, especially in postmenopausal women and the elderly.) 3.2: Pre-clinical performance assessment under various environmental conditions

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Development of a portable and high-performance multimodal POC platform for monitoring osteoporosis disease progression (e.g., P1NP/CTx)** • Develop immobilization protocols of nanobodies on the nanopatterned electrodes with enhanced alignment • Develop highly sensitive and selective nanobody-based biosensors capable of detecting multiplexed biomarkers for monitoring of osteoporosis disease progression • Integrate multimodal detection technologies • Validation of prototype POC kit in the laboratory using biological samples as pre-clinical testing data collection, and analysis • Disease-specific assays optimized for multiplexed detection • Real-world performance assessment under various environmental conditions **Intellectual Property (IP) & Technology Transfer** • Patent applications for immunoassay technologies, biosensor platforms, and integrated diagnostic devices • Licensing and commercialization strategies for industry adoption • Technology transfer to diagnostic companies for market deployment
Expected R&R of Korean SME Partner	**Develop and establish a protocol for the synthesis of multiple high affinity nanobodies** **Develop nanopatterned electrodes** • Design for manufacturability: define critical dimensions, pattern pitch/height, allowable tolerances, and wafer-level layout for high yield • Process flow establishment: adhesion layer → metal deposition → nanopatterning → lift-off/etch • Thin-film deposition: sputtering/evaporation of Au/Pt/Ti/Cr layers; control of thickness for electrochemical stability • Etch & surface engineering: RIE/wet etch optimization; surface activation (O_2 plasma/UV-ozone) for consistent chemistry • Yield & reproducibility study: DOE for key parameters (pressure, dose, temp/pressure, etch time); Cp/Cpk tracking across lots

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**A portable and high-performance multimodal POC platform for monitoring osteoporosis disease progression** • Established synthesis protocols of well-defined nanobodies for biosensor applications • Established semiconductor fabrication methods for mass-produced and reproducible nanopatterned electrodes • Established immobilization protocols of nanobodies on the nanopatterned electrodes • Developed highly sensitive and selective nanobody-based biosensors capable of detecting multiplexed biomarkers for monitoring osteoporosis disease progression • Integrated multimodal detection technologies • Validation of prototype POC kit in the laboratory using biological samples. **Field evaluation and pre-clinical validation** • Real-world performance assessment under various environmental conditions **Cross-validated assay panels for osteoporosis disease** • Disease-specific assays optimized for multiplexed detection • Data on analytical performance, such as limit of detection, cross-reactivity, and reproducibility, is shared across sites for highly keen validation **Joint intellectual property and technology commercialization** • Technology licensing and commercialization planning through Korean SMEs and MGB • Filing of joint patents based on co-developed technologies **Strengthened Korea-MGB R&D Collaboration** • Established a long-term research network between academic, clinical, and industrial partners in both countries • Publication of joint peer-reviewed articles (SCI-indexed journal paper) and conference presentations to disseminate research findings **Commercialization potential** • **Product:** Portable cartridge–reader POC kit with recurring consumable sales (single-use cartridges/reagents). • **Market fit:** Fast duplex monitoring of **P1NP/CTx** in decentralized settings (clinics, health-check centers, primary care). • **Scale-up:** Wafer-level nanopatterned electrodes enable **mass production** and cost competitiveness via Korean SME/diagnostic partners. **Broader industrial/economic impact** • **Industry upgrading:** Strengthens the domestic value chain in **nanobody reagents + semiconductor biosensors + POC device integration**. • **Healthcare impact:** Enables earlier/continuous monitoring to support better treatment decisions and reduce fracture-related burden. • **Economic spillover:** Creates high-value IP and commercialization pathways, supporting SME growth and export potential through Korea–MGB collaboration.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization